THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 5–7: **Arithmetic Progressions, Triangles, Coordinate Geometry** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(c)** 40	First find $d = 3$
2	**(a)** 4	$AE = k \times AD$
3	**(b)** 24	Find the hypotenuse
4	**(c)** 24	$n = \frac{l-a}{d} + 1$
5	**(a)** 5	Perimeter ratio = side ratio
6	**(d)** 8	Midpoint
7	**(d)** 1665	AP of two-digit multiples
8	**(a)** 6	$h^2 = pq$
9	**(c)** 4	$\frac{x_1+x_2+x_3}{3}$
10	**(c)** 623	Use S_n
11	**(b)** 36	Similar triangles
12	**(b)** 5	Distance from centre
13	**(d)** 23	Middle term $= \frac{a+l}{2}$
14	**(b)** 100	Half-diagonals are legs
15	**(a)** $\frac{1}{2}$	$x = \frac{kc+a}{k+1} = m$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$a_n = a + (n-1)d$: $127 = 7 + (n-1)(6)$ (1); $n - 1 = 20$ (1); $n = 21$, so it is the 21st term (1).	1+1+1
17	$\frac{AD}{AB} = \frac{AE}{AC}$ (1); $AE = \frac{3 \times 10}{5} = 6$ cm (1); $EC = 10 - 6 = 4$ cm (1).	1+1+1
18	$(6)^2 + (y-2)^2 = 100$ (1); $(y-2)^2 = 64$ (1); $y = 2 \pm 8$, i.e. $y = 10$ or -6 (1).	1+1+1
19	$a = 14$, $d = 6$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{18} = \frac{18}{2}[28 + 17(6)] = 1170$ (1).	1+1+1
20	Ratio of sides = ratio of perimeters $= 1 : 2$ (1); side $= 8$ cm (1); ratio of areas $= 1 : 4$ (1).	1+1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$a + 2d = 19$, $a + 7d = 54$ (2); subtracting, $d = 7$, $a = 5$ (2); $a_{15} = 103$ (1).	2+2+1
22	$\frac{AD}{AB} = \frac{AE}{AC}$ (1); $AE = \frac{4 \times 18}{9} = 8$ cm (1); $EC = 18 - 8 = 10$ cm (1).	1+1+1
23	$AB^2 = 16$, $BC^2 = 13$, $CA^2 = 13$ (2); two sides are equal (1); hence isosceles (1).	2+1+1
24	$a_1 = 5$, $a_{20} = 62$ (1); $d = 3$ (1); $S_{20} = \frac{20}{2}(5 + 62) = 670$ (1).	1+1+1
	OR: $\frac{PQ}{AB} = \frac{QR}{BC} = \frac{RP}{CA} = 2$ (1); $QR = 8$ cm (1); $RP = 10$ cm (1).	1+1+1
25	Section formula $\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right)$ (1); $x = \frac{2(8)+3(-2)}{5} = 2$ (1); $y = 7$ (1).	1+1+1
	OR: $S_n = \frac{n}{2}[2a + (n-1)d]$: $185 = \frac{10}{2}[10 + 9d]$ (1); $d = 3$ (1); $a_{10} = 32$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	$S_n = \frac{n}{2}[2(54) + (n-1)(-4)] = 360$ (1); form quadratic in n (2); $n = 10$ (the other root is rejected) (2).	1+2+2
	OR: Draw a horizontal line from the top of the shorter pole: right triangle with legs 9 and 12 (2); $x^2 = 81 + 144$ (1); $x = 15$ m (2).	2+1+2
27	Let ratio $m : n$; $y = \frac{m(4)+n(-5)}{m+n} = 0$ (2); $m : n = \frac{5}{4} : 1$, i.e. 5:4 (1); point $x = \frac{5(6)+4(-3)}{9} = 2$ (2).	2+1+2
	OR: AP with $a = 45$, $l = 25$, $n = 11$ (1); $S_n = \frac{11}{2}(45 + 25) = 385$ cm (2); middle term $= \frac{45+25}{2} = 35$ cm (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 5–7: **Arithmetic Progressions, Triangles, Coordinate Geometry** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(a)** 4	$a = a_p - (p-1)d$
2	**(a)** 13	Right triangle with legs d and $a-b$
3	**(a)** 10	Area $= 0$
4	**(d)** 52	First find $d = 6$
5	**(a)** 9	$AE = k \times AD$
6	**(a)** 30	Find the hypotenuse
7	**(c)** 31	$n = \frac{l-a}{d} + 1$
8	**(c)** 7	Perimeter ratio = side ratio
9	**(a)** 6	Midpoint
10	**(d)** 1188	AP of two-digit multiples
11	**(b)** 6	$h^2 = pq$
12	**(a)** 3	$\frac{x_1+x_2+x_3}{3}$
13	**(b)** 640	Use S_n
14	**(c)** 64	Similar triangles
15	**(d)** 13	Distance from centre

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Diagonals bisect each other: midpoint of AC is $\left(\frac{11}{2}, \frac{7}{2}\right)$ (2); this is also the midpoint of BD (1); $D = (5, 5)$ (2).	2+1+2
17	First term 4, last term 96 (1); $n = 24$, sum $= 1200$ (1).	1+1
18	Midpoint theorem: $DE \parallel BC$, $DE = \frac{1}{2}BC = 4$ cm (2); areas in ratio $(1/2)^2 = 1:4$ (1).	2+1
19	Centroid $= \left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$ (1); $= (6, 4)$ (2).	1+2
20	$a_n = a + (n-1)d$: $79 = 11 + (n-1)(4)$ (1); $n - 1 = 17$ (1); $n = 18$, so it is the 18th term (1).	1+1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Altitude^2 = product of the segments (1); $h^2 = 4 \times 9 = 36$ (1); $h = 6$ cm (1).	1+1+1
22	Area $= \frac{1}{2}\lvert x_1(y_2-y_3) + x_2(y_3-y_1) + x_3(y_1-y_2)\rvert$ (1); $= \frac{1}{2}\lvert -12 + 12 + 32\rvert$ (1); $= 16$ sq. units (1).	1+1+1
23	$a + d = 2$, $a + 6d = 32$ (2); subtracting, $d = 6$, $a = -4$ (2); $a_{12} = 62$ (1).	2+2+1
24	$\frac{AD}{AB} = \frac{AE}{AC}$ (1); $AE = \frac{2 \times 27}{9} = 6$ cm (1); $EC = 27 - 6 = 21$ cm (1).	1+1+1
	OR: $AB^2 = 16$, $BC^2 = 20$, $CA^2 = 20$ (2); two sides are equal (1); hence isosceles (1).	2+1+1
25	$a_1 = 4$, $a_{15} = 74$ (1); $d = 5$ (1); $S_{15} = \frac{15}{2}(4 + 74) = 585$ (1).	1+1+1
	OR: $\frac{PQ}{AB} = \frac{QR}{BC} = \frac{RP}{CA} = 3$ (1); $QR = 18$ cm (1); $RP = 21$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Diagonals bisect at right angles: half-diagonals 6 and 8 (1); $s^2 = 6^2 + 8^2 = 100$ (1); side $= \sqrt{100}$ cm (1).	1+1+1
	OR: Let the point be $(x, 0)$; $(x-1)^2 + 25 = (x-7)^2 + 9$ (2); $x = \frac{8}{3}$ (2); point $(\frac{8}{3}, 0)$ (1).	2+2+1
27	$S_n = \frac{n}{2}[2(12) + (n-1)(6)] = 714$ (1); form quadratic in n (2); $n = 14$ (the other root is rejected) (2).	1+2+2
	OR: Draw a horizontal line from the top of the shorter pole: right triangle with legs 16 and 12 (2); $x^2 = 256 + 144$ (1); $x = 20$ m (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 5–7: **Arithmetic Progressions, Triangles, Coordinate Geometry** | Class 10 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** 28	Middle term $= \frac{a+l}{2}$
2	**(b)** 169	Half-diagonals are legs
3	**(d)** $\frac{1}{2}$	$x = \frac{kc+a}{k+1} = m$
4	**(a)** 10	$a = a_p - (p-1)d$
5	**(a)** 10	Right triangle with legs d and $a-b$
6	**(a)** 9	Area $= 0$
7	**(d)** 39	First find $d = 2$
8	**(c)** 12	$AE = k \times AD$
9	**(d)** 36	Find the hypotenuse
10	**(a)** 17	$n = \frac{l-a}{d} + 1$
11	**(d)** 9	Perimeter ratio = side ratio
12	**(d)** 9	Midpoint
13	**(c)** 945	AP of two-digit multiples
14	**(a)** 6	$h^2 = pq$
15	**(c)** 5	$\frac{x_1+x_2+x_3}{3}$

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$\frac{AD}{AB} = \frac{AE}{AC}$ (1); $AE = \frac{4 \times 20}{10} = 8$ cm (1); $EC = 20 - 8 = 12$ cm (1).	1+1+1
17	$(6)^2 + (y+1)^2 = 100$ (1); $(y+1)^2 = 64$ (1); $y = -1 \pm 8$, i.e. $y = 7$ or -9 (1).	1+1+1
18	$a = -12$, $d = 4$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{26} = \frac{26}{2}[-24 + 25(4)] = 988$ (1).	1+1+1
19	Ratio of sides = ratio of perimeters $= 1:3$ (1); side $= 15$ cm (1); ratio of areas $= 1:9$ (1).	1+1+1
20	Diagonals bisect each other: midpoint of AC is $\left(3, \frac{5}{2}\right)$ (2); this is also the midpoint of BD (1); $D = (2, 4)$ (2).	2+1+2

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Section formula $\left(\frac{mx_2+nx_1}{m+n}, \frac{my_2+ny_1}{m+n}\right)$ (1); $x = \frac{3(7)+1(1)}{4} = \frac{11}{2}$ (1); $y = 5$ (1).	1+1+1
22	$S_n = \frac{n}{2}[2a + (n-1)d]$: $360 = \frac{12}{2}[16 + 11d]$ (1); $d = 4$ (1); $a_{12} = 52$ (1).	1+1+1
23	Altitude^2 = product of the segments (1); $h^2 = 2 \times 8 = 16$ (1); $h = 4$ cm (1).	1+1+1
24	Area $= \frac{1}{2}\lvert x_1(y_2-y_3) + x_2(y_3-y_1) + x_3(y_1-y_2)\rvert$ (1); $= \frac{1}{2}\lvert -32 - 4 - 36\rvert$ (1); $= 36$ sq. units (1).	1+1+1
	OR: $a + 3d = 3$, $a + 8d = -12$ (2); subtracting, $d = -3$, $a = 12$ (2); $a_{14} = -27$ (1).	2+2+1
25	$\frac{AD}{AB} = \frac{AE}{AC}$ (1); $AE = \frac{6 \times 18}{9} = 12$ cm (1); $EC = 18 - 12 = 6$ cm (1).	1+1+1
	OR: $AB^2 = 16$, $BC^2 = 20$, $CA^2 = 20$ (2); two sides are equal (1); hence isosceles (1).	2+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Let ratio $m:n$; $y = \frac{m(7)+n(-8)}{m+n} = 0$ (2); $m:n = \frac{8}{7}:1$, i.e. 8:7 (1); point $x = \frac{8(4)+7(1)}{15} = \frac{13}{5}$ (2).	2+1+2
	OR: AP with $a = 60$, $l = 30$, $n = 13$ (1); $S_n = \frac{13}{2}(60 + 30) = 585$ cm (2); middle term $= \frac{60+30}{2} = 45$ cm (2).	1+2+2
27	Diagonals bisect at right angles: half-diagonals 5 and 12 (1); $s^2 = 5^2 + 12^2 = 169$ (1); side $= \sqrt{169}$ cm (1).	1+1+1
	OR: Let the point be $(x, 0)$; $(x+3)^2 + 4 = (x-5)^2 + 36$ (2); $x = 3$ (2); point $(3, 0)$ (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.